

PS3 Chunk

Abhishu Karki

The `summary_table()` function is designed to make it easy to create summary tables for data analysis. It is based on a revised version of the table-formatting function I developed when I reviewed my midterm exam. In many research settings, analysts repeatedly calculate basic statistics such as sample size, means, and standard deviations for different groups. This function automates that process by allowing the user to specify a grouping variable and a numeric variable, and then produces a clean, presentation-ready table using tidyverse syntax and `kableExtra`.

Function

```
penguins <- penguins

library(tidyverse)
library(knitr)
library(kableExtra)

summary_table <- function(data, group_var, numeric_var,
                           caption = NULL, digits = 1) {

  summary_df <- data |>
    filter(
      !is.na({{ group_var }}),
      !is.na({{ numeric_var }})
    ) |>
    group_by({{ group_var }}) |>
    summarise(
      N = n(),
      Mean = round(mean({{ numeric_var }}), digits),
      SD = round(sd({{ numeric_var }}), digits),
      .groups = "drop"
```

```

) |>
  rename(Group = {{ group_var }})

summary_df |>
  kbl(
    caption = caption,
    align = c("l", "c", "c", "c"),
    booktabs = TRUE
  ) |>
  kable_styling(
    position = "center",
    latex_options = "hold_position"
  ) |>
  row_spec(0, bold = TRUE, background = "#2E86AB", color = "#F2F0EF") |>
  column_spec(1, bold = TRUE)
}

```

demonstration

```

speciestable <- summary_table( penguins, species, body_mass, "Mean Body Mass
  ↪ of Penguins by Species (grams)" )

gendertable <- summary_table( penguins, sex, flipper_len, "Mean Flipper
  ↪ Length by Sex (millimeters)" )

islandtable <- summary_table( penguins, island , bill_len, "Mean Bill Length
  ↪ by Island (millimeters)" )

```

speciestable

Table 1: Mean Body Mass of Penguins by Species (grams)

Group	N	Mean	SD
Adelie	151	3700.7	458.6
Chinstrap	68	3733.1	384.3
Gentoo	123	5076.0	504.1

gendertable

Table 2: Mean Flipper Length by Sex (millimeters)

Group	N	Mean	SD
female	165	197.4	12.5
male	168	204.5	14.5

islandtable

Table 3: Mean Bill Length by Island (millimeters)

Group	N	Mean	SD
Biscoe	167	45.3	4.8
Dream	124	44.2	6.0
Torgersen	51	39.0	3.0

Thank You